

Ex 1: DL en O.

① $f_1(x) = \frac{\ln(1+x)}{1+x}$ $n=4$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + x^4 \varepsilon_1(x)$$

$$1+x = 1+x + x^4 \varepsilon_2(x)$$

$$\frac{\ln(1+x)}{1+x} = \frac{x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + x^4 \varepsilon_1(x)}{1+x + x^4 \varepsilon_2(x)}$$

$$\frac{1}{3} - \frac{1}{2}$$

$\begin{array}{r} x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \\ - x + x^2 \\ \hline -\frac{3}{2}x^2 + \frac{x^3}{3} - \frac{x^4}{4} \\ - \frac{3}{2}x^2 - \frac{3}{2}x^3 \\ \hline \frac{5}{6}x^3 + x^4 \\ \frac{5}{6}x^4 + \frac{5}{6}x^4 \\ \hline \frac{1}{6}x^4 \end{array}$	$\begin{array}{r} 1+x \\ x - \frac{3}{2}x^2 + \frac{5}{6}x^3 + \frac{1}{6}x^4 \end{array}$
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$$\frac{\ln(1+x)}{1+x} = x - \frac{3}{2}x^2 + \frac{5}{6}x^3 + \frac{1}{6}x^4 + x^4 \varepsilon(x).$$

② $f_2(x) = \tan^2 x$ $n=3$

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + x^5 \varepsilon(x)$$

$$\tan^2 x = \left(x + \frac{x^3}{3} + x^3 \varepsilon(x)\right) \left(x + \frac{x^3}{3} + x^3 \varepsilon(x)\right)$$

$$= x^2 + x^3 \varepsilon(x)$$

$$g) f_3(x) = (1+x)^{\frac{1}{3}} \quad n=3.$$

$$\text{on a } (1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \frac{\alpha(\alpha-1)(\alpha-2)}{3!} x^3 + x^3 \varepsilon(n)$$

$$\begin{aligned} \alpha &= \frac{1}{3} \\ (1+x)^{\frac{1}{3}} &= 1 + \frac{1}{3}x + \frac{\frac{1}{3}(\frac{1}{3}-1)}{2} x^2 + \frac{\frac{1}{3}(\frac{1}{3}-1)(\frac{1}{3}-2)}{6} x^3 + x^3 \varepsilon(n) \\ &= 1 + \frac{1}{3}x + \frac{-2}{18} x^2 + \frac{5}{81} x^3 + x^3 \varepsilon(n) \end{aligned}$$

$$d) f_4(x) = \frac{1 - \cosh(x)}{\sinh(x)} \quad n=2.$$

$$\cosh(x) = 1 + \frac{x^2}{2} + x^2 \varepsilon(n)$$

$$\sinh(x) = x + x^2 \varepsilon(n)$$

$$1 - \cosh(x) = -\frac{x^2}{2} + x^2 \varepsilon(n)$$

$$f_4(x) = \frac{-\frac{x^2}{2} + x^2 \varepsilon_1(n)}{x + x^2 \varepsilon_2(n)} = -\frac{x}{2} + x \varepsilon(n)$$

$$e) f_5(x) = e^x - \sqrt{1+2x} \quad n=2$$

$$e^x = 1 + x + \frac{x^2}{2} + x^2 \varepsilon(n)$$

$$\sqrt{1+2x} = 1 + \frac{x}{2} - \frac{1}{8} x^2 + x^2 \varepsilon_1(n)$$

$$\sqrt{1+2x} = 1 + \frac{2x}{2} - \frac{1}{8} (4x^2) + (2x)^2 \varepsilon_1(n)$$

$$= 1 + x - \frac{1}{2} x^2 + x^2 \varepsilon(n)$$

$$\begin{aligned} f_5(x) &= 1 + x + \frac{x^2}{2} - \left(1 + x - \frac{1}{2} x^2 + x^2 \varepsilon(n) \right) \\ &= x^2 + x^2 \varepsilon(n) \end{aligned}$$

$$f) f_6(x) = (1+2x)^{\frac{1}{1+x}} \quad n=3.$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^3 \varepsilon(n)$$

$$(1+2x)^{\frac{1}{1+x}} =$$

$$(1+2x)^{\frac{1}{1+x}} = \exp\left(\frac{1}{1+x} \ln(1+2x)\right)$$

$$\begin{aligned} \ln(1+2x) &= 2x - \frac{2x^2}{2} + \frac{(2x)^3}{3} + x^3 \varepsilon(n) \\ &= 2x - 2x^2 + \frac{8x^3}{3} + x^3 \varepsilon(n) \end{aligned}$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + x^3 \varepsilon(n)$$

$$\frac{1}{1+x} \ln(1+2x) = (1-x+x^2-x^3) \left(2x - 2x^2 + \frac{8x^3}{3}\right)$$

$$= 2x - 2x^2 + \frac{8x^3}{3} - \frac{2x^2 + 2x^3 + 2x^3}{3} + x^3 \varepsilon(n)$$

$$= 2x - 4x^2 + \frac{20}{3}x^3 + x^3 \varepsilon(n) \xrightarrow{n \rightarrow \infty} 0$$

$$\exp\left(\frac{1}{1+x} \ln(1+2x)\right) = 1 + 2x - 4x^2 + \frac{20}{3}x^3 + \frac{4x^2 - 16x^3}{2} +$$

$$= 1 + 2x - 4x^2 + \frac{20}{3}x^3 + 2x^2 - 8x^3 + \frac{4}{3}x^3 + x^3 \varepsilon(n)$$

$$= 1 + 2x - 2x^2 + x^3 \varepsilon(n)$$

$$g) f_7(n) = (1 + \sin n)^{\frac{1}{n}} \quad n = 2$$

$$f_7(x) = \exp\left(\frac{1}{x} \ln(1 + \sin n)\right)$$

$$1 + \sin n = 1 + x + \frac{x^3}{6} + n^2 \varepsilon(n)$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + n^2 \varepsilon(n)$$

$$\frac{1}{x} \ln(1+x) = \frac{x - \frac{x^2}{2} + \frac{x^3}{3}}{x} = \underbrace{1 - \frac{x}{2} + \frac{x^2}{3}}_u + n^2 \varepsilon(n)$$

$$\exp\left(\frac{1}{x} \ln(1+x)\right) = e \left(\exp\left(\underbrace{-\frac{x}{2} + \frac{x^2}{3}}_u\right) \right)$$

$$u = \left(-\frac{x}{2} + \frac{x^2}{3}\right)^2 = \frac{x^2}{4} + n^2 \varepsilon(n)$$

$$f_7 = e \cdot \left(1 - \frac{x}{2} + \frac{x^2}{3} + \frac{x^2}{8} \right) + n^2 \varepsilon(n)$$

$$= e \left(1 - \frac{x}{2} + \frac{11}{24} x^2 + n^2 \varepsilon(n) \right)$$

$$R) f_8(n) = \frac{\arctan x}{\arcsin n} \quad n = 2$$

$$f_8(n) \neq \frac{1}{\arccos n}$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots + (-1)^n \frac{x^{2n+1}}{2n+1} + n^{2n+3} \varepsilon(n)$$

$$\arcsin n = n + \frac{1}{2} \frac{n^3}{3}$$

$\begin{array}{r} x - \frac{2x^3}{3} + \frac{x^5}{5} \\ \hline x + \frac{1}{6} x^3 \\ \hline -\frac{1}{2} x^3 + \frac{x^5}{8} \\ \hline -\frac{1}{2} x^3 - \frac{1}{12} x^5 \\ \hline \frac{1}{6} x^5 \end{array}$	$\begin{array}{r} x + \frac{1}{6} x^3 \\ \hline 1 - \frac{1}{2} x^2 \end{array}$
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$$f_8(n) = 1 - \frac{1}{2} n^2 + n^2 \varepsilon(n)$$

Ex2:

1^{ère} méthode: Taylor

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + (x-a)^2 \varepsilon(x)$$

$$a = \frac{\pi}{4}$$

$$f\left(\frac{\pi}{4}\right) = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$$

$$f'(x) = -\sin x \Rightarrow f'\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$$

$$f''(x) = -\cos x \Rightarrow f''\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$$

$$\cos x = \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right) + \frac{\sqrt{2}}{4} \left(x - \frac{\pi}{4}\right)^2 + \left(x - \frac{\pi}{4}\right)^2 \varepsilon(x)$$

2^{ème} méthode:

$$f_1(x) = \cos x \quad a = \frac{\pi}{4} \quad n = 2$$

$$\text{on pose } h = x - a \quad \begin{array}{l} x \rightarrow \frac{\pi}{4} \\ h \rightarrow 0 \end{array} \quad x = h + \frac{\pi}{4}$$

$$f_1(x) = \cos\left(h + \frac{\pi}{4}\right) = \cos h \cos \frac{\pi}{4} - \sin h \sin \frac{\pi}{4}$$

$$= \frac{\sqrt{2}}{2} \cos h - \frac{\sqrt{2}}{2} \sin h$$

$$= \frac{\sqrt{2}}{2} \left(1 - \frac{h^2}{2} - h + h^2 \varepsilon(h)\right)$$

$$= \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \left(x - \frac{\pi}{4}\right) - \frac{\sqrt{2}}{4} \left(x - \frac{\pi}{4}\right)^2 + \left(x - \frac{\pi}{4}\right)^2 \varepsilon(x)$$

$$b) f_2(x) = \arctan x \quad a = 1 \quad n = 3$$

Taylor:

$$f_2(x) = \arctan(1) + (\arctan(1))' (x-1) + \frac{(\arctan(1))''}{2} (x-1)^2 + \frac{(\arctan(1))^{(3)}}{3!} (x-1)^3 + (x-1)^3 \varepsilon(x)$$

$$\arctan 1 = \frac{\pi}{4}$$

$$(\arctan x)' = \frac{1}{1+x^2} \rightarrow f_2'(1) = \frac{1}{2}$$

$$(\arctan x)'' = \frac{-2x}{(1+x^2)^2} \rightarrow f_2''(1) = -\frac{2}{4} = -\frac{1}{2}$$

$$(\arctan x)^{(3)} = \frac{-2(1+x^2)^2 + 4x(1+x^2)2x}{(1+x^2)^4} =$$

$$= \frac{-2(1+x^2) + 8x^2}{(1+x^2)^3} = \frac{-2 + 8x^2}{8} = \frac{+6}{8} = \frac{+3}{4}$$

$$f_2(x) = \frac{\pi}{4} + \frac{1}{2}(x-1) - \frac{1}{4}(x-1)^2 + \frac{1}{12}(x-1)^3 + (x-1)^3 \varepsilon(x)$$

$$c) f_3(x) = \frac{1+x}{1-x} \quad a = 2 \quad n = 2$$

$$h = x - 2 \Rightarrow x = h + 2$$

$$f_3(x) = \frac{1+h+2}{1-h-2} = \frac{h+3}{-(h+1)} = -\frac{h+3}{h+1}$$

$$\cancel{h+3} \times \cancel{h+1}$$

Ex 3:

$$\begin{aligned}
 * \quad & \lim_{x \rightarrow 0} \frac{1}{\sin^2 x} - \frac{1}{x^2} \\
 &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x}{x^2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{(x - \sin x)(x + \sin x)}{x^2 \sin^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{(\cancel{x} - \cancel{x} + \frac{x^3}{6} + x^4 \varepsilon(x))(\cancel{x} + \cancel{x} - \frac{x^3}{6} + x^4 \varepsilon(x))}{x^2 (x^2 + x^2 \varepsilon(x))} \\
 &= \lim_{x \rightarrow 0} \frac{\frac{2x^4}{6} + x^4 \varepsilon(x)}{x^4 + x^4 \varepsilon(x)} = \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 * \quad & \lim_{x \rightarrow 0} \frac{e^x - e^{\sin x}}{x^3} \\
 \sin x &= x - \frac{x^3}{6} + x^3 \varepsilon(x) \\
 e^{\sin x} &= e^{x - \frac{x^3}{6} + x^3 \varepsilon(x)} \xrightarrow{x \rightarrow 0} 0 \\
 &= 1 + x - \frac{x^3}{6} + \frac{(x - \frac{x^3}{6})^2}{2} + \frac{(x - \frac{x^3}{6})^3}{6} + x^3 \varepsilon(x) \\
 &= 1 + x - \frac{x^3}{6} + \frac{x^2}{2} + \frac{x^3}{6} + x^3 \varepsilon(x) \\
 e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + x^3 \varepsilon(x) \\
 e^x - e^{\sin x} &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} - 1 - x + \frac{x^3}{6} + x^3 \varepsilon(x) \\
 &= \frac{x^3}{6} + x^3 \varepsilon(x) \\
 \lim_{x \rightarrow 0} \frac{e^x - e^{\sin x}}{x^3} &= \frac{\frac{x^3}{6} + x^3 \varepsilon(x)}{x^3} = \frac{1}{6}
 \end{aligned}$$

$$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{\sinh x - x \cosh x}$$

$$\sin x = x - \frac{x^3}{6} + x^3 \varepsilon(x)$$

$$x \cos x = x \left(1 - \frac{x^2}{2} + x^2 \varepsilon(x) \right) = x - \frac{x^3}{2} + x^3 \varepsilon(x)$$

$$\sinh(x) = x + \frac{x^3}{6} + x^3 \varepsilon(x)$$

$$x \cosh(x) = x \left(1 + \frac{x^2}{2} + x^2 \varepsilon(x) \right) = x + \frac{x^3}{2} + x^3 \varepsilon(x)$$

$$\lim_{x \rightarrow 0} \frac{\sin x - x \cos x}{\sinh(x) - x \cosh(x)} = \lim_{x \rightarrow 0} \frac{x - \frac{x^3}{6} - x + \frac{x^3}{2} + x^3 \varepsilon(x)}{x + \frac{x^3}{6} - x - \frac{x^3}{2} + x^3 \varepsilon(x)}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{3} x^3 + x^3 \varepsilon(x)}{-\frac{1}{3} x^3 + x^3 \varepsilon(x)} = -1.$$

Ex 3:

(1) $x \mapsto e^{\frac{x}{1+x}}$ à l'ache 3 en 0

$$\frac{x}{1+x} \rightarrow 0 \quad x \rightarrow 0$$

$$e^{\frac{x}{1+x}} = 1 + \frac{x}{1+x} + \frac{\left(\frac{x}{1+x}\right)^2}{2} + \frac{\left(\frac{x}{1+x}\right)^3}{6} + x^3 \varepsilon(x)$$

$$= 1 + \frac{x}{1+x} + \frac{3x^2(1+x)}{2(1+x)^2} + \frac{x^3}{6(1+x)^3} + x^3 \varepsilon(x)$$

$$= \frac{6 + 18x + 18x^2 + 6x^3 + 6x + 12x^2 + 6x^3 + 3x^2(3x^3) + x^3}{6(1+x)^3}$$

$$= \frac{16x^3 + 33x^2 + 24x + 6}{6 + 18x + 18x^2 + 6x^3}$$

15
18
33

Division euclidienne

$$\begin{array}{r|l}
 6 + 24x + 33x^2 + 16x^3 & 6 + 18x + 18x^2 + 6x^3 \\
 \hline
 6 + 18x + 18x^2 + 6x^3 & 1 + x - \frac{1}{2}x^2 + \frac{1}{6}x^3 \\
 \hline
 6x + 15x^2 + 10x^3 & \\
 6x + 18x^2 + 18x^3 + 6x^4 & \\
 \hline
 -3x^2 - 8x^3 + 6x^4 & \\
 -3x^2 - 9x^3 & \\
 \hline
 x^3 &
 \end{array}$$

$$DL_3 \text{ de } f(x) = e^{\frac{x}{1+x}} = 1 + x - \frac{1}{2}x^2 + \frac{1}{6}x^3 + x^3 \varepsilon(x)$$

② DL_3 de $f(x) = \sqrt{\cos 2x}$.

$$\begin{aligned}
 \cos 2x &= 1 + \frac{(2x)^2}{2} + x^3 \varepsilon(x) \\
 &= 1 + 2x^2 + x^3 \varepsilon(x)
 \end{aligned}$$

$$\sqrt{\cos 2x} = \sqrt{1 + (2x^2 + x^3 \varepsilon(x))} =$$

$$\begin{aligned}
 \sqrt{1+x} &= 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3(1.3)}{2 \cdot 4 \cdot 6 \cdot 2} \\
 &= 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + x^3 \varepsilon(x)
 \end{aligned}$$

$$\begin{aligned}
 \sqrt{\cos 2x} &= 1 + \frac{2x^2 + x^3 \varepsilon(x)}{2} + x^3 \varepsilon(x) \\
 &= 1 + x^2 + x^3 \varepsilon(x)
 \end{aligned}$$

$$(3) f(x) = e^{\frac{x}{1+x}} \sqrt{\cos 2x}.$$

a) DL₃ de f .

ona DL₃ de $e^{\frac{x}{1+x}}$ et DL₃ $\sqrt{\cos 2x}$

$$f(x) = \left(1+x - \frac{1}{2}x^2 + \frac{1}{6}x^3 + x^3\varepsilon(x)\right) \left(1+x^2 + x^3\varepsilon(x)\right)$$

$$= 1+x^2+x+x^3 - \frac{1}{2}x^2 + \frac{1}{6}x^3 + x^3\varepsilon(x)$$

$$= 1 + \frac{x^2}{2} + x + \frac{7}{6}x^3 + x^3\varepsilon(x)$$

$$= 1+x + \frac{x^2}{2} + \frac{7}{6}x^3 + x^3\varepsilon(x)$$

b) Tangente $y = f(0) + f'(0)x$.

par DL $y = 1+x$.

$$c) f(x) - y = \frac{x^2}{2} + \frac{7}{6}x^3$$

le signe est le signe $f''(0) = 1 > 0$

alors φ_f est au dessus de la tangente

Ex 5:

$$f(n) = \frac{x^2}{1+n} e^{\frac{1}{n}}$$

① DL₂ de $\frac{f(n)}{n}$.

on pose $x = \frac{1}{n} \rightarrow \frac{f(n)}{n} = x f\left(\frac{1}{x}\right)$

$$f\left(\frac{1}{x}\right) = \frac{\left(\frac{1}{x}\right)^2}{1 + \frac{1}{x}} e^x = \frac{x}{x^2(x+1)} e^x$$

$$= \frac{1}{x^3 + x^2} e^x$$

$$= \frac{1}{x(x+1)} e^x = \left(\frac{1}{x} - \frac{1}{x+1}\right) e^x$$

$$= \left(\frac{1}{x} - \left(1 - x + x^2 - x^3 + x^3 \varepsilon(x)\right)\right) e^x$$

$$x f(x) = (1 - x + x^2 - x^3 + x^3 \varepsilon(x)) e^x$$

$$e^x = 1 + x + \frac{x^2}{2} + x^2 \varepsilon(x)$$

$$x f(x) = (1 - x + x^2 + x^2 \varepsilon(x)) \left(1 + x + \frac{x^2}{2} + x^2 \varepsilon(x)\right)$$

$$= 1 + x + \frac{x^2}{2} - x - x^2 + x^2 + x^2 \varepsilon(x)$$

$$= 1 + \frac{x^2}{2}$$

$$\frac{f(n)}{n} = 1 + \frac{1}{2n^2} + \frac{1}{n^2} \varepsilon\left(\frac{1}{n}\right)$$

$$2) \lim_{n \rightarrow \infty} \frac{f(n)}{n} = 1. \Rightarrow y=1 \text{ est une asyn.}$$

$$3) \lim_{n \rightarrow \infty} \frac{f(n)}{n} = 1 = \frac{1}{2n^2} > 0$$

$\Rightarrow$ ~~asymptote~~ Cf est au dessus de l'asymptote $y=1$.

Ex 6:

$$]-\frac{\pi}{4}, \frac{\pi}{4}[\setminus \{0\}$$

$$(1) DL_3 \text{ de } R(n) = e^{\sqrt{1+\sin n}} - e.$$

$$\sqrt{1+\sin n} = \sqrt{1+n+\frac{n^2}{8} + n^2 \varepsilon(n)} = 1 + \frac{n}{2} - \frac{n^2}{8} + \frac{1}{16}n^3 + \frac{3}{8}n^2 \varepsilon(n)$$

$$e^{\sqrt{1+\sin n}} = e^{1 + \frac{n}{2} - \frac{n^2}{8} + n^2 \varepsilon(n)} = e \cdot e^{\left(\frac{n}{2} - \frac{n^2}{8} + \frac{1}{16}n^3 + \frac{3}{8}n^2 \varepsilon(n)\right)}$$

$$R(n) = e \left(1 + \frac{n}{2} - \frac{n^2}{8} + \frac{1}{16}n^3 + \frac{3}{8}n^2 \varepsilon(n) \right) - e$$

$$= \frac{n}{2} + n^2 \varepsilon(n)$$

$$\lim_{n \rightarrow 0} \frac{R(n)}{n} = \frac{\frac{n}{2}}{n} = \frac{1}{2}.$$

$$g^3(n) = g(0) + g'(0) \cdot \frac{n}{2} + g''(0) \frac{n^2}{2}$$

Ex 6:

① D_L de $R(x) = e^{\sqrt{1+\sin x}} - e$ $x \in]-\frac{\pi}{4}, \frac{\pi}{4}[\setminus \{0\}$

$$\sqrt{1+\sin x} = \sqrt{1+x+x^2\epsilon(x)} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{1}{16}x^3 + x^3\epsilon(x)$$

$$\begin{aligned} e^{\sqrt{1+\sin x}} &= e^{1 + \frac{x}{2} - \frac{x^2}{8} + \frac{1}{16}x^3 + x^3\epsilon(x)} \\ &= e \cdot e^{\frac{x}{2} - \frac{x^2}{8} + \frac{1}{16}x^3 + x^3\epsilon(x)} \end{aligned}$$

$$\begin{aligned} R(x) &= e \left(1 + \frac{x}{2} - \frac{x^2}{8} + \frac{1}{16}x^3 + \frac{x^2}{8} - \frac{x^3}{16} + x^3\epsilon(x) \right) - e \\ &= e \frac{x}{2} + x^3\epsilon(x). \end{aligned}$$

② $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{R(x)}{\tan x} = \lim_{x \rightarrow 0} \frac{\frac{ex}{2} + x^3\epsilon(x)}{x + \frac{1}{3}x^3 + x^3\epsilon(x)}$

$$= \lim_{x \rightarrow 0} \frac{x \left(\frac{1}{2} + x^2\epsilon(x) \right)}{x \left(1 + \frac{1}{3}x^2 + x^2\epsilon(x) \right)} = \frac{e}{2}.$$

③
$$g(x) = \begin{cases} f(x) & \text{si } x \in]-\frac{\pi}{4}, \frac{\pi}{4}[\setminus \{0\} \\ \frac{1}{2} & \text{si } x = 0 \end{cases}$$